

```

1  #include <stdio.h>
2
3  int main()
4  {
5      printf("Enter number of processes: ");
6      int n;
7      scanf("%d", &n);
8
9      int burstTime[10];
10     int processId[10];
11
12     for(int i = 0; i < n; i++)
13     {
14         processId[i] = i + 1;
15         printf("Enter burst time of process P%d: ", i + 1);
16         scanf("%d", &burstTime[i]);
17     }
18
19     // Sort processes by burst time
20     for(int i = 0; i < n - 1; i++)
21     {
22         for(int j = i + 1; j < n; j++)
23         {
24             if(burstTime[j] < burstTime[i])
25             {
26                 int temp = burstTime[i];
27                 burstTime[i] = burstTime[j];
28                 burstTime[j] = temp;
29
30                 temp = processId[i];
31                 processId[i] = processId[j];
32                 processId[j] = temp;
33             }
34         }
35     }
36
37     int waitingTime[10];
38     int turnaroundTime[10];
39
40     int totalWaitingTime = 0;
41     int totalTurnaroundTime = 0;
42
43     for(int i = 0; i < n; i++)
44     {
45         if(i == 0)
46         {
47             waitingTime[i] = 0;
48         }
49         else
50         {
51             waitingTime[i] = waitingTime[i - 1] + burstTime[i - 1];
52         }
53
54         turnaroundTime[i] = waitingTime[i] + burstTime[i];
55
56         totalWaitingTime += waitingTime[i];
57         totalTurnaroundTime += turnaroundTime[i];
58     }
59
60     printf("\nSJF Scheduling\n");
61     printf("-----\n");
62     printf("PID\tBT\tWT\tTAT\n");
63     printf("-----\n");
64
65     for(int i = 0; i < n; i++)
66     {
67         printf("P%d\t%d\t%d\t%d\n",
68             processId[i],
69             burstTime[i],
70             waitingTime[i],
71             turnaroundTime[i]);
72     }
73
74     float avgWaitingTime = (float) totalWaitingTime / n;
75     float avgTurnaroundTime = (float) totalTurnaroundTime / n;
76
77     printf("-----\n");
78     printf("Average Waiting Time = %.2f\n", avgWaitingTime);
79     printf("Average Turnaround Time = %.2f\n", avgTurnaroundTime);
80
81     return 0;
82 }
83

```